Effect of cadmium sulphide on Electrical and Optical properties of Polyaniline

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By

Mohammad Hafeez Khan
Enrollment no-16DTPH006

Under Supervision of

Dr. Biswajit Saha

Department of Physics

NATIONAL INSTITUTE OF TECHNOLOGY,

AGARTALA-799046, INDIA

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Approval Sheet

This dissertation entitled ***"Effects of Cadmium sulphide on electrical and optical properties of Polyaniline"*** by Mohammad Hafeez Khan (BT-MT Eight semester candidate) is approved for the BT degree at the 8th semester of BT-MT programme in Engineering Physics.

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Signature of Supervisor

Dr.Biswajit Saha

Department of Physics

NIT Agartala

8th June 2019

Preface

The project entitled ***"Effect of Cadmium Sulphide on Electrical and Optical properties on polyaniline"*** is carried out as a part and partial fulfillment of BT-MT degree in Physics from NIT Agartala. The project is initiated in the eight semester of BT-MT course in the department of Physics of NIT Agartala.

The project is aimed for synthesis of organic-inorganic nanocomposites. The structural properties of Nano-composite particles have been investigated using X-ray diffraction data obtained from diffraction pattern. The optical properties have been studied using UV-visible absorption spectrum and the electrical property analysis using the I-V pattern of the nano composites used.

Effective application of X-ray diffraction and I-V are the analytical tools to understanding of crystallographic nature and the conductive property of the material respectively. Therefore, I have gone through such a project to have a better understanding of structural as well as the electrical properties. UV-visible absorption spectra also have confirmed its conductive phase.

Mohammad Hafeez Khan

8th June 2020

Department of Physics

National Institute of Technology, Agartala

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Chapter 1

1. Introduction

1.1 Introduction to conducting Polymers

Polymers are large organic molecules consisting of long carbon chain in which monomer units repeat it to form a long chain. In polymer atoms are connected to each other by covalent bonds which may form a conjugated system. Mostly the polymers are insulators as there are no free electrons. But there are many polymers which are electrically conductive if charge carries are created in a polymer. The conducting polymer are extended pi-conjugated system in which there is a alternating double and single bonds which helps in charge transportation once a charge is created in polymer and makes the doping process a bit easier. The charge movement takes place with the help of resonance through the series of conjugated bonds, these types of polymers are known conducting polymers. Conducting polymers are often typically intrinsically conductive or at least semiconductors. The process of charge creation can be done by various processes like by oxidation, reduction and protanation of polymer during synthesis. Due to their light weight and simple fabrication techniques, conducting polymers are of great application and replacing metals in several areas of application.

The earliest reported organic conductive material was polyaniline in 1862 by Henry Letheby, after that tetraiodopyrrole with high conductivity was found and in 1977 it was discovered that polyacetylene can be oxidized with halogens to produce conducting materials. In 1970 that a first intrinsic conducting polymer was produced by Shirakawa, Heegar and MacDiarmid for which they were awarded a noble prize in 2000. After that many efforts have been made to develop the properties of conducting polymer to meet the needs in various technologies.

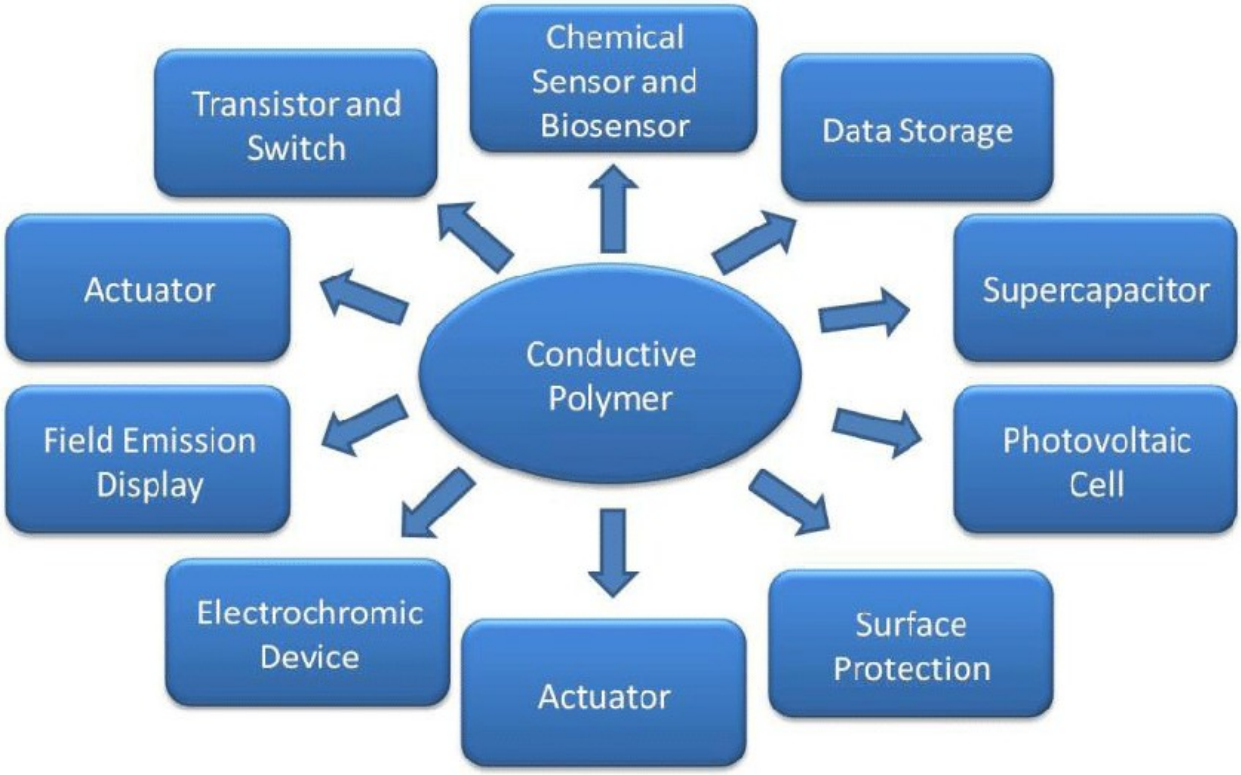


Fig. shows various applications of conducting polymers.

1.2 Conducting Polymer

The pioneering work of Alan Heeger, Alan G. MacDiarmid and Hideki Shirakawa during the 1970's which culminated in the Nobel Prize in Chemistry for unveiling the unprecedented merits of the class of conducting polymers, was the mile stone for identifying the potential of this class of materials in a wide variety of technological applications. The surprisingdiscovery of the metallic properties in the linear conjugated organic polymer, polyacetylene and the additional surprises related to the possibilities of tuning the electrical conducting properties of this polymer from insulating to superconducting, based on the doping strategies adopted, triggered a surge of worldwide research activity to identify more categories of polymers with enhanced electrically conducting properties. A variety of approaches for the synthesis of conducting polymers was soon identified and based on both chemical and electrochemical routes different types of conducting polymers with impressive characteristics could be realized. The prime members of this group of conducting polymers with potential application prospects include polyacetylene,

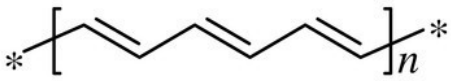
polyaniline, polypyrrole, polythiophene, polyphenylenevinylene and their group of substituted polymers, co-polymers and hybrid polymers.

Among those polymers endowed with electrical conducting properties suitable for applications in active devices, polyaniline (PANI) is one of the highly sought-after materials, because of its unique material characteristics, and exceptionally good environmental stability, not observed in any of the other polymers. Polyaniline is highly stable in air and is an eco-friendly material, possessing quite fascinating electronic and optoelectronic properties with excellent application prospects in the emerging technological fields.

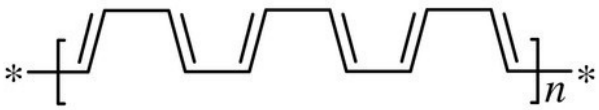
1.3 *Types of Conductive Polymer*

Polyacetylene

Polyacetylene (PA) was the first conducting polymer to be synthesized late back in 1958. PA is organic polymer in which $(C_2H_2)_n$ unit repeat itself. It can be synthesized in many ways; simplest way to synthesize polyacetylene is by direct polymerization of acetylene by passing acetylene gas through the solution of Zeigler Natta catalyst in hydrocarbon solvent which results in the precipitation of trans-Polyacetylene. Conductivity for doped cis is more than the trans-polyacetylene. I_2, Br_2, Li, Na, AsF_5 can alkali metals are used as doping materials. Due to its insolubility, infusibility and poor environmental stability it is not highly recommended for technological applications.



trans-polyacetylene



cis-polyacetylene

Fig. **1.1** cis and trans-polyacetylene

Polypyrrole

Polypyrrole have been one of important conducting polymer due to its high electrical conductivity, high chemical and thermal stability, good mechanical properties and it can be easily synthesized. PPy is of great applications such as in light weight batteries, electronic devices, and chemical sensors, also used in microwave fabrication of multiwalled carbon nano tubes and chromatographic stationary phases. Polypyrrole can be synthesized by electrochemical or chemical polymerization. BF_4^- and ClO_4^- can be used as doping materials.

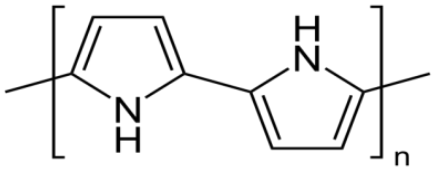


Fig. 1.2 Polypyrrole Polymer

Polyaniline

Polyaniline (PANI) is a conducting polymer containing an alternate benzene rings and nitrogen atoms in its unique structure. The nitrogen atoms in PANI can exist either as imine or amine. It has received a great attention as its Electrical conductivity of PANI can be controlled by varying the pH of the medium and is environmentally stable. Different oxidation states of polyaniline can be generated such as Leucoemeraldine (fully reduced), Emeraldine, Protonated Emeraldine, Nigraniline and Pernigraniline (fully oxidized) by doping with oxidants at different pH values. Among these Protonated Emeraldine which is moderately oxidized and is the only conducting form of polyaniline. Polyaniline are of great interest in physics and chemistry due to the present of -NH-group in its backbone. The major applications of polyamine are manufacturing of printed circuit board, for electromagnetic shielding of electronic circuits, corrosion protection, and flexible rods, for antistatic coating and in photo electro chemical cell. Polyaniline can be synthesizes by electrochemical or chemical oxidation.

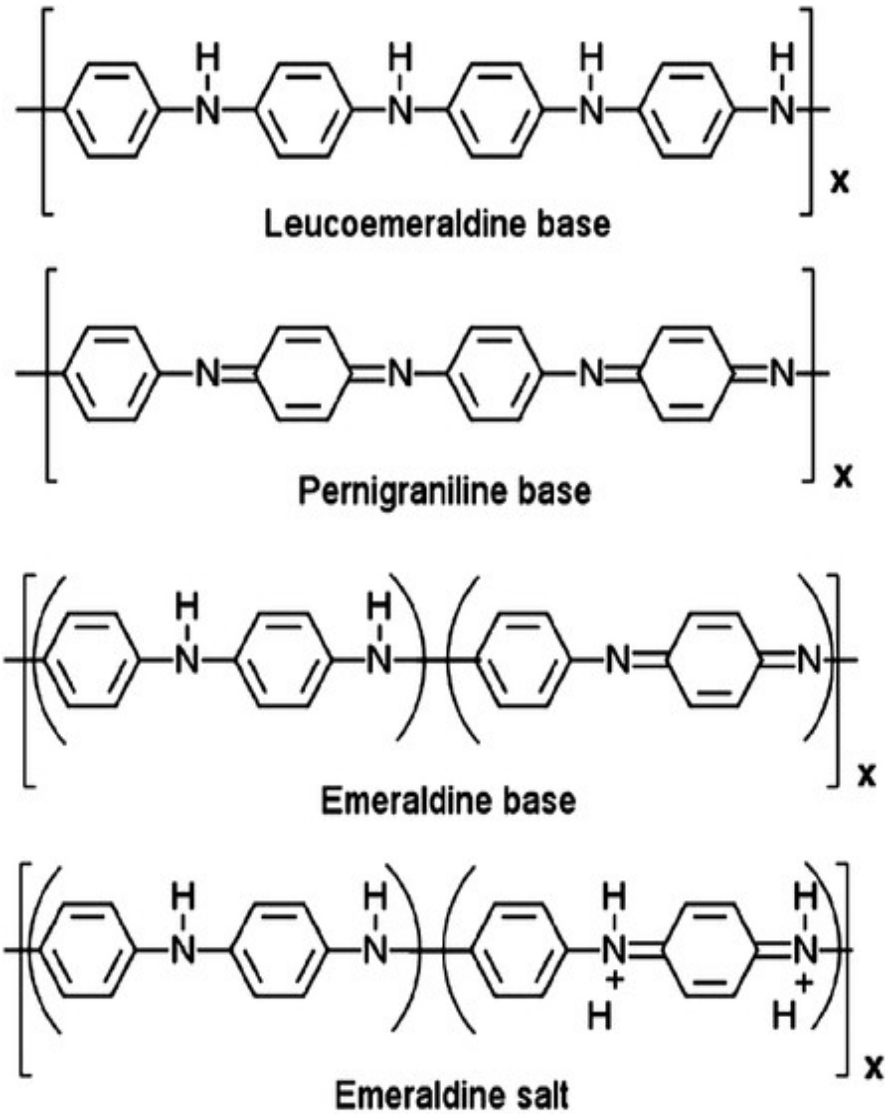


Fig. 1.3 Different oxidation states of poylanine Polymer

Polythiophenes

Polythiophene (PT) is polymerized thiophenes, a sulfur hetrocycle. They can be conducting when electrons are added or removed from the conjugated π-orbitals via doping. PT has been found of great interest in research because it has better thermal stability, good mechanical strength, high environmental stability and less band energy gap. It also shows better electronic and optical properties. Due to these properties PT has been used in lots of industrial areas like

field effect transistor, solar cell, electro chromic devices, biosensors and electroluminescent materials.

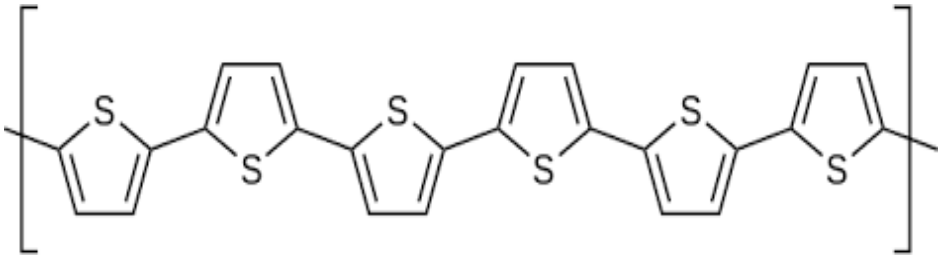


Fig. 1.4 Polythiophene

Polyphenylene Vinylene

Poly (p-phenylene vinylene) (PPV or polyphenylene vinylene) is a conducting polymer. PPV is the only polymer among conducting polymer which can be easily synthesized in good purity and high molecular weight. Moreover it can be easily doped to form electrically conductive material. Its physical and electronic properties can be altered by inclusion of functional side group. PPV is capable of showing small optical band gap, electroluminescence, and shows greater thermal and mechanical strength. This polymer has been of great use in the field of electronic devices such as organic light emitting diodes, photovoltaic devices, and organic solar cells.

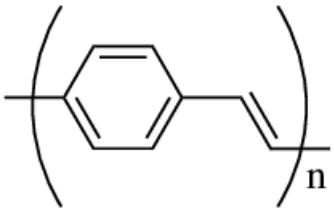


Fig. 1.5 Polyvinylene Vinylene

1.4 Charge carriers in conductive polymers

Initially polymers are insulating but they can be conducting if they are doped. Doping of polymer means introducing charge carrier in polymer chain by oxidation or reduction. Electrical properties of conductive polymers have been of great interest since their discovery. One of the factors on which electrical properties depend is charge carriers responsible for the electrical conduction. Mainly three type of charge carriers are involved in conduction mechanism of polymer namely polarons, bipolarons and soliton.

Polarons

When a conducting polymer consisting of a conjugated backbone is oxidized by an oxidizing agent, as result a cation and a free radical is formed in polymer chain. The radical and cation are coupled with each other through resonance along the length of polymer chain. A polymer chain having a cation and radical is called polaron. Similarly it can be anion and radical if conductive polymer is reduced.

Bipolaron

If the polarons formed after the oxidation are further oxidized, oxidizing agent removes the radical and two cation are formed in the polymer chain which can propagate along the length of carbon chain through resonance with π-bonds. These types of structure are called bipolarons. If the conducting polymer is highly doped then two polarons combine to form a bipolarons.

Soliton

Conducting polymer with degenerate ground state like polyacetylene has slightly different mechanism. It can exist in two different forms cis and Trans, which have exactly same energy. Due to two fold degenerate in some conducting polymer, the charged cation is not bound to each other by a higher energy configuration and can freely separate along the chain and they start moving from each other along the chain. The two structure cis and trans which differ from one another by the exchange of alternating single and double carbon- carbon double bond along the chain. This degeneracy leads to the formation of so called soliton and sometimes they are neutral.

1.5 Applications of conductive polymers

By virtue of their light weight and greater ease of fabrication, have replaced and are continuing to replace metals in several areas of applications. Conducting polymers have been of great interest and have found enormous application in science and technology due their unique electrical and optical properties. Some of applications have been discussed below.

Batteries

Batteries are the first application of conducting polymers which drew its attention in the industrial world. Light weight and rechargeable batteries can be prepared using either polyaniline or polyprrole as electrodes. Solid state batteries can also prepared using Polymer conductors.

Light Emitting Diodes

Conducting polymers have been used in the synthesis of Organic light emitting diodes (OLEDs). Due to their unique electrical properties thin conducting polymers layers have been used as hole injection layer in opto electronic devices, which improves the charge transport mechanism and have improved the LEDs. A complete range (PLEDs) has been produced over the entire visible region.

Corrosion Protection

Coating of conducting polymers such as polyprrole and Polyaniline has been used for the corrosion protection of ferrous and non-ferrous metals.

Sensors

Conducting polymer materials have been widely used in manufacturing different kind of sensors. Biosensor such as glucose biosensor can be fabricated. These biosensors have found huge application such as in food testing, medical clinics and in agricultural industries. Gas sensors are also prepared using conducting polymers which can be used in detection of harmful gas. Humidity sensors have been prepared which have application in industrial and medical field.

Solar Cells

Solar cell based on conducting polymer have been developed so far that they are capable of replacing the silicon based solar cell along with their advantages such as low cost, light weight, high flexible strength and simple processing.

Electro chromic Display

Elecrochromic device is another important application which involves electrochemical doping and undoping. Doping process changes the electronic structure of conducting polymer which results in chromatic effects, they have been of great importance in display technologies.

Supercapacitors

Supercapacitors or ultra capacitor are electrochemical capacitors and their family has emerged of great use in energy storage and memory devices. Due to their low cost and ease in processing Conducting polymers has been used in synthesis of electrode in these supercapacitors that have improved their capacitive energy density.

1.6 Cadmium Sulphide Nanoparticles

Wayne Shenton, Dietmar Pum, Uwe B. Sleytr and Stephen Mann were the first to synthesis cadmium sulphide nanoparticles by biological method. The cadmium sulfide is an important II-IV group element semiconductor (Eg=2.42eV (515nm) at room temperature) with many excellent physical and chemical properties. This has promising application in multiple technical fields including photochemical catalysis, gas sensor, detectors for laser and infrared Cadmium sulfide is a solid hexagonal or cubic crystal. Cadmium sulfide is an important n-type semiconductor with a direct band gap 2.42eV, the photoconduction and electroluminescent properties of cadmium sulfide have been applied in manufacturing a verity of consumer goods.



Fig. 1.6 Cadmium Sulphide powder

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Chapter 2

2. Review of literature

2.1 Organic Inorganic Nanocomposites

The development of polymer nanocomposites is currently one of the most active areas in the field of nanotechnology and Nanoscience. These polymer nanocomposites are mixture of inorganic and organic compounds which are of great application in electronics devices, coatings, electro chromic devices, optics, fuels and solar cells. Combination of such polymers and metal or semiconductor nanoparticles results in creation of new type of conducting polymer-metal or semiconductor nanocomposites. Adding a doping agent to these polymer nanocomposites can change their electrical properties quite significantly.

Organic conducting polymer itself does not possess excellent electronic, mechanical, thermal, optical, chemical and physical properties which are required for various applications, but by the addition of inorganic nanomaterials to these conducting polymers enhance their various properties like electrical conductivity, thermal stability, optical properties and one can also tune these properties by varying the amount inorganic compound in these conducting polymers.

During last few years there are growing interests on composite materials, mainly organic and inorganic hybrid materials. Particular attentions have been paid on nano composites. Many nano composites such as Polyaniline/NiO, PANI/ZnS, PEDOT/ZnO, Polypyrrole/TiO_2, Ppy/CuO, PEDOT/NiO etc. are being explored recently with excellent physical properties. There are parallel developments in synthesis processes of organic and inorganic nano composites.

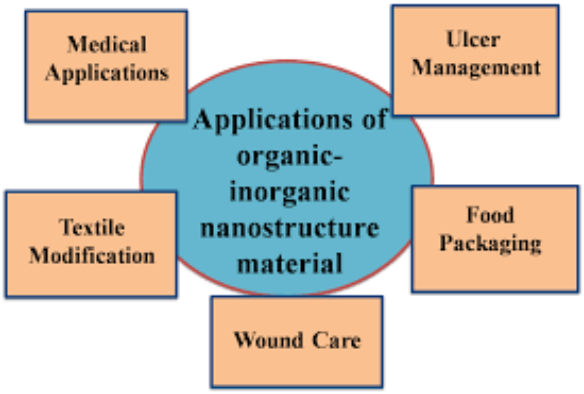


Fig. 2.1 Applications of organic-inorganic nanocomposites.

2.2 Review of Synthesis of Inorganic-Organic Polymer nanocomposites

Some recent techniques used in synthesis of inorganic and organic nanocomposites have been reviewed and discussed below.

Synthesis of CdS nanoparticles have been reported by B.Srinivasa Rao *et al* (2011). In which CdS nanoparticles were prepared by the colloidal chemical precipitation method using cadmium acetate ($CdCH_3COOH$), sodium sulfide (Na_2S) as starting compounds and using thiophenol as capping agent. The stoichiometric solution was taken in a burette and was added in drops with continuous sitting until a precipitate of CdS was formed. After complete precipitation, the solution in conical flask was constantly stirred for about 20 hours. Then the precipitates were filtered out. The samples were calcined at 300°C/2hrs in vacuum.

Syed Khasim*et al* (2018) reported the synthesis of PANI/CdS nanocomposites films. CdS nanoparticles were synthesized using co-precipitation method and EDTA was used as a stabilizing agent. In this process cadmium nitrate was dissolved in mixed solution of deionized water and ethanol taken in equal volume and stirred in round bottom flask at 80^{0}C. A sulfur source of $Na_2S.9H_2O$ was also separately dissolved in deionized water-ethanol solvents and was added drop-by-drop to the above solution and the mixture as stirred for couple of minutes resulting in a yellow solution. To this solution ethanol was added followed by the addition of solution containing EDTA drop wise with constant vigorous stirring for couple of hours at 80^{0}C. The resulted mixture was eventually cooled to room temperature and final dispersion was centrifuged. The obtained precipitation was washed several times with water and treated with alcohol to eliminate traces of EDTA, impurities and original reactants. After

washing wet precipitate was dried. Then PANI was synthesized by chemical oxidative polymerization of aniline I the presence of ammonium peroxy disulphate as an oxidizing agent. PANI/CdS composite films were prepared by simple casting technique. Solution of CdS nanoparticles was prepared by dispersing in dimethyl propylene urea under sonication and the synthesized PANI was added under vigorous stirring to get good dispersion. Finally the resulting suspension was filtered through syringe filter of 50 microns and the suspension was cast onto the clean glass plate and dried under 20 mm Hg at 323 K for twelve hours to obtain freestanding films of PANI/CdS nanocomposites.

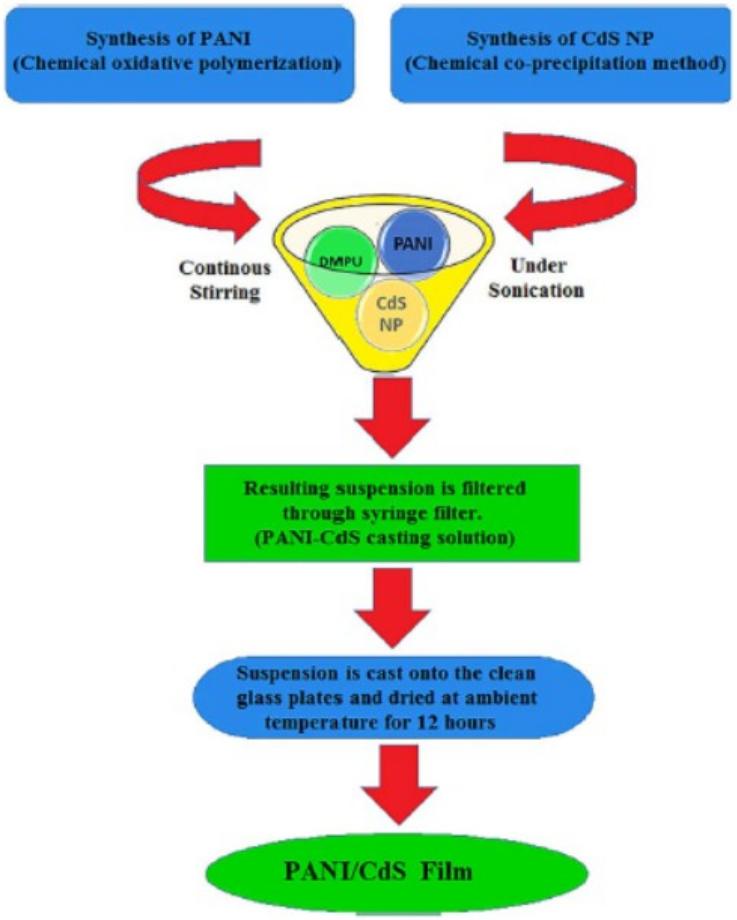


Fig. 2.2 Schematic representation of composite synthesis of PANI/CdS Film.

Polyaniline–cadmium sulfide nanocomposites have been reported by M. Ghoswami*et al* (2015) using in-situ as follows. Cadmium acetate was mixed with distilled

water and sodium sulfide was also mixed distilled water. Ammonium peroxodisulfate (APS) were mixed with distilled water and kept this solution in refrigerator. Na_2S is mixed drop wise in the cadmium acetate solution; a yellow precipitate of cadmium sulfide (CdS) is obtained. Then a double distilled aniline and acetic acid is mixed to this solution with magnetic stirring. Ice cooled APS solution is added to this solution taken in an ice bath. A green colored solution is obtained after two hours of stirring. The solution was kept in refrigerator at rest for 24 h to complete the polymerization process and then this solution is centrifuged at 10,000 rpm for 30 min. The solid mass obtained was washed with acetone, ethanol and double distilled water to remove monomer, oligomer and excess of oxidant until the filtrate turned colorless.

ZnS nano particles embedded in polyaniline synthesized by chemical technique have been reported by Kousik Dutta*et al* (2008). In which 1.75 g of zinc perchlorate ($Zn(ClO_4)_2$) was dissolved in 250 ml of deionized water and then stirred for 1 h at room temperature. Aqueous Solution of thioglycerol (12 mM) was mixed drop wise with the aqueous solution of zinc perchlorate under constant stirring. The pH of the resultant solution was kept at 11 with the addition of NaOH aqueous solution. After 45 min Na_2S solution was mixed slowly with the above reaction mixture and stirred for 1 h at 80 ◦ C and to cool to room temperature. The solution was dialyzed against water and mixed with 2-propanol to make as a precipitate. Finally the resultant product was washed several times with methanol to remove the unwanted impurities and dried in a vacuum oven at 70 ◦ C for 6 h. The required quantity of ZnS nanoparticles was ultrasonically dispersed in 50 ml deionized water. Aniline monomer of known volume was slowly added into the dispersion under sonication at room temperature. Then the aqueous solution of APS maintaining a aniline:APS ratio of 1:1.25 was added drop wise. After few hours the resulting solution was turned into green color which indicates the formation of polyaniline. The solution was then kept under sonication for about 6 h to ensure complete polymerization. Finally the resulting green residues were centrifuged. The resulting nanocomposites were washed thoroughly with distilled water for several times.

2.3 Properties of polymer Inorganic/Metal nanocomposites

Organic-Inorganic metal oxide compounds have several important properties. Some of the properties have been reviewed and discussed here below.

DC Conductivity with various temperatures of NiO/Polyaniline synthesized by in-situ polymerization method with different weight percentage of Ni (10, 20, 30, 40 and 50wt %) has been reported by Sharanabasamma M Ambalgi*et al* (2016). The DC conductivity of the samples was measured in the temperature range of 30^0-190^0C and it was observed that DC conductivity increases with the increasing concentration of nickel oxide concentration. Experimental results were supported by the equation and theory of three-dimensional Mott's variable range hopping model.

$$\sigma(T) = \sigma_o \exp\left[-\left(\frac{T_o}{T}\right)^{1/4}\right]$$

Where σ is the conductivity, T is the temperature and σ_0 is the conductivity at characteristic temperature T_0. The increase in conductivity of the composites may be due the formation of more number of polarons. Composite having 50 wt% of PANI/NiO2 was reporting the higher conductivity among all concentration of NiO and than pure PANI.

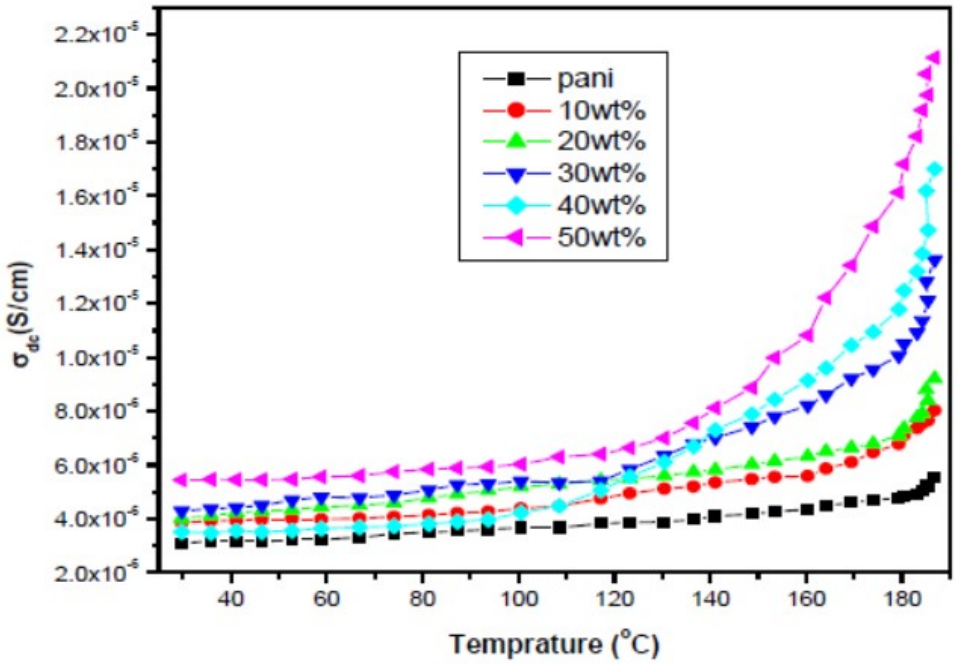
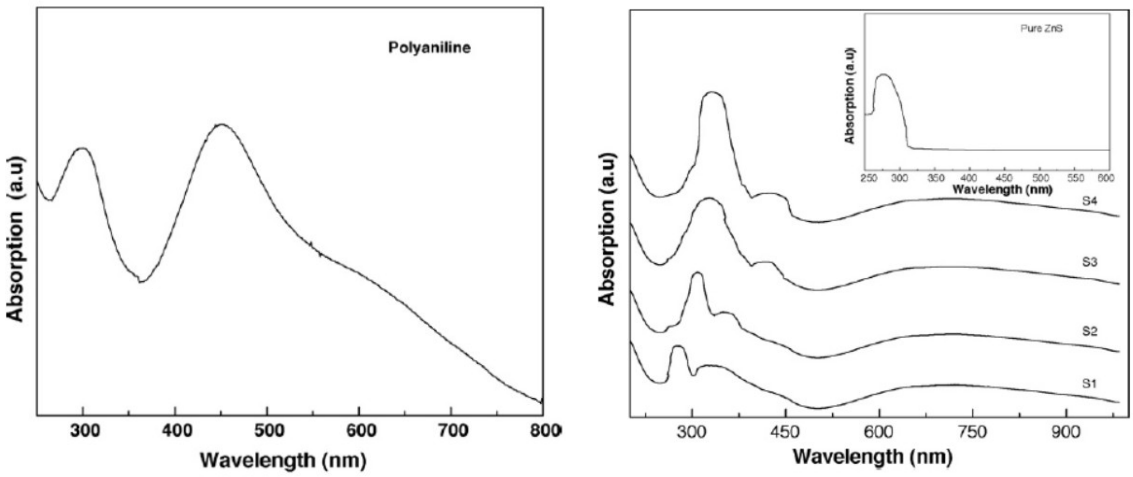


Fig. 2.3 Conductivity of different weighted NiO/PANI with temperature.

Kousik Dutta*et al* (2008) reported the optical and electrical properties of ZnS nanoparticles embedded in polyaniline. UV-vis absorption spectra were studied for different weight percentage (71S4, 78S3, 87S2, and 95S1 wt% of aniline) of nanocomposites. A prominent absorption band appears at wavelength of about 270nm. The optical absorption edge is blue-shifted from the absorption edge (345nm) of bulk ZnS. The absorption spectra for (S1, S2, S3 and S4) consist of three absorption bands. Polyaniline reveals absorption at 300nm and a strong peak around 450nm followed by a wider shoulder around 650nm. The optical band gap E_g is calculated using tauc plot. The estimated band gap of pure ZnS in nanocomposites is 4.59 ev. The calculated band gaps of ZnS in nanocomposites were found to be 4.50, 4.0, 3.85, 3.75 ev for S1, S2, S3 and S4 respectively. This shows that the band gap of nanocomposites decreases as the wt% of ZnS increases in the nanocomposites.



(a) **(b)**

Fig. 2.4 (a) UV-vis absorption spectrum of polyaniline. (b) UV–vis absorption spectra of the nanocomposites sample S1–S4. The inset represents the same for pure ZnS.

The Ac conductivity σ(ω) as a function of temperature were studied for S4. The frequency independent behavior of σ(ω) at low frequency established the absence of electrode polarization. The conductivity starts to increase after certain characteristic onset frequency ω_0. The onset frequency shifts to higher frequency region with increasing temperature. The σ(ω) spectrum within the highest domain is found at the lowest temperature (143K).

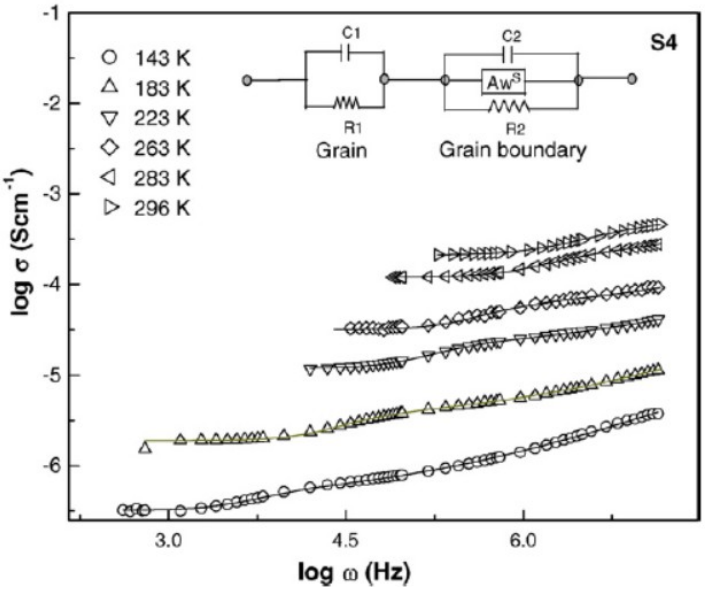
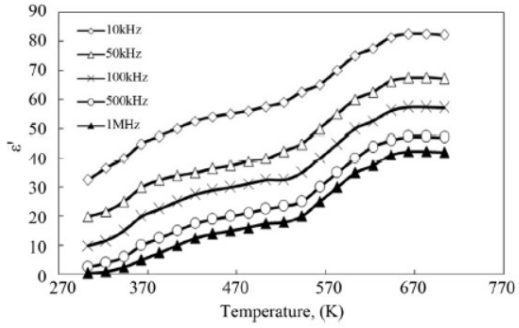
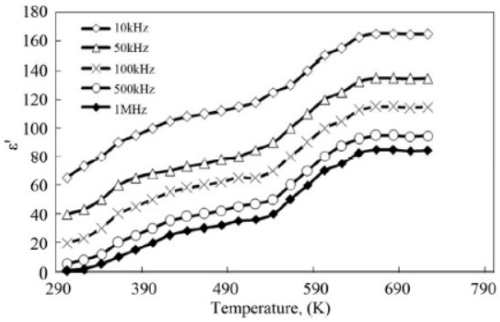


Fig. **2.5** shows frequency dependence of conductivity at different temperatures for the sample S4.

Safenaz M. Reda *et al* (2012) reported the synthesis of PANI/Ag nanocomposites and studied their electrical and optical properties. One of the interesting properties reported in this research was the temperature dependence of dielectric permittivity. At all the frequencies the ϵ' value for PANI/Ag nanocomposites is higher than that of pure one. Temperature dependence of dielectric constants of PANI and PANI/Ag in the frequency range of 303-723K was studied. Dielectric constant of PANI/Ag was found to be higher than that of pure PANI. It was seen that the incorporation of Ag enhanced the dielectric constant of PANI/Ag nanocomposites about three times of that recorded for pure PANI under the same conditions. The better dielectric constant of the PANI/Ag nanocomposites indicates their better ability to store electric potential energy under the influence of alternative electric field.



(a)



(b)

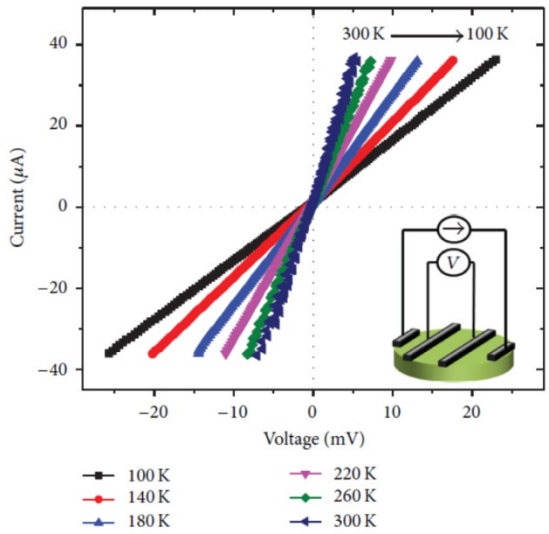
Fig. 2.6 Effect of temperature on dielectric constant of (a) PANI and (b) PANI/Ag nanocomposites.

Mohd. Khalid *et al* (2013) reported the synthesis and the electrical properties of polyaniline nanotubes doped with different sulfonic acid. Self –assembled polyaniline (PANI)

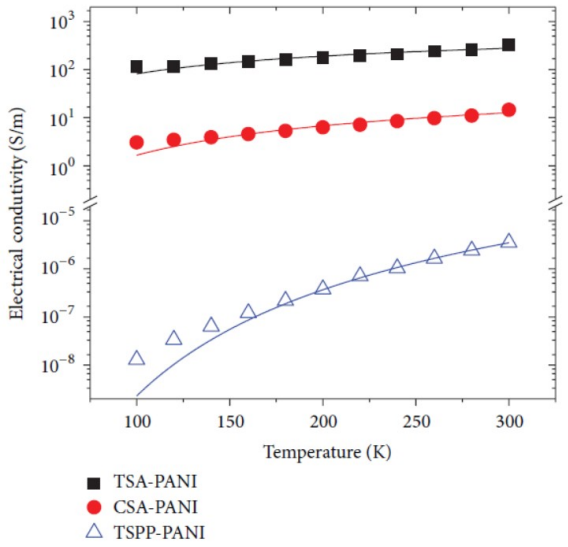
nanotubes were prepared in the presence of three different sulfonic acid as dopant, namely p-toulenesulfonic acid (TSA), camphorsulfonic acid (CSA), tetrakis(4-sulfonatophenyl)porphyrin (TSSP), by oxidative polymerization using ammonium peroxydisulphate as the oxidant. Their electrical properties were studied with different temperatures. I vs V curves of TSA-PANI samples are shown in Fig. 2.7 (a). The linear trend was observed for all samples measured in this work. The conductivity can be calculated from the slope. The electrical conductivities are shown in Fig 7(b) as a function of temperature. All samples showed a clear reduction of the conductivity by lowering the temperature, that is, less than one order of magnitude for TSA and CSA doped PANI nanotubes and about two orders of magnitude for TSPP doped PANI nanotubes. The values for conductivity at room temperature are 30 S/m for TSA-PANI, 15 S/m for CSA-PANI, and 3×10^{-6} S/m for TSPP-PANI. TSA and CSA-PANI are altleast, seven times more conductive than undoped PANI, making these materials suitable for electrical application. The dependency of the conductivity with temperature follows an activation behavior that can be described by the exponential equation.

$$\sigma(T) = \sigma_o \exp\left[-\left(\frac{T_o}{T}\right)^{1/4}\right]$$

Where σ_0 is constant and, T_0 is Mott characteristic temperature of system.



(a)



(b)

Fig. 2.7 (a) I×V curves at different temperatures, obtained by four probe measurements, for TSA-PANI sample. The inset illustrates the sample contacts and the electrical measurement. **(b)** Conductivity of the three doped samples as a function of the temperature.

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Chapter 3

3. Synthesis Process

Based on the literature survey and recent trends in Physics of sensors, it appears that organic conducting polymers are of significant potential in exploring their applications in science and technology. Keeping this in mind we set our objective to work with a polymer based electronic materials that can be employed for low cost sensor materials and functional materials for electronic devices. Accordingly, a work plan is framed to execute a project work with polyaniline. The work plan would be executed on the Effects of CdS on Electrical and Optical properties of Polyaniline. Before that the synthesis of the Polyaniline will be done, and the proposed synthesis with the materials required for synthesis process is illustrated here.

3.1 Materials

Aniline monomer, methanol, cadmium acetate (Cd (CH_3 COO) $_2$.$2H_2O$), thiourea (CH_4N_2S), ammonium per sulfate (APS) and hydrochloric acid (Merck) were used in the chemical synthesis process. All the chemical reagents used in this experiment were of analytical grade and used without further purification.

3.2 Synthesis of PANI powder

PANI powder was prepared by following the chemical polymerization technique. Briefly, 0.1M aniline was dissolved in 1.0M aqueous solution of HCl in an ice bath maintained at the temperature of 0-5 ^{0}C.The oxidant solution of APS was mixed drop wise in the solution and stirring was continued for 4 h to complete the polymerization of aniline. The dark green color edemeraldine salt precipitate was obtained after filtration, which was then washed repeatedly with distilled water until the filtrate become colorless. Finally, the precipitate was washed by methanol to remove the oligomeric impurities and dried at temperature of 70 ^{0}C for 8 h on a hot plate to obtain the pure PANI powder.

3.3 Synthesis of nano crystalline CdS

Nano crystalline CdS powder has been synthesized by a sol–gel process in which cadmium acetate (Cd (CH_3 COO) $_2$.$2H_2O$) has been used as a parent source for Cd and thiourea (CH_4N_2S) has been used as a source of S. For experiment, 3.198 g of cadmium acetate and 1.824 g of thiourea was added into 40 ml of methanol and mixture was stirred vigorously at 60^{0}C for 1 h. After gel formation heating was stopped and the solution was stirred continuously until we get a yellow powder. The powder was dried in a furnace at 300 ^{0}C for 30 min to get the nano crystalline CdS powder.

3.4 Synthesis of PANI/CdS nanocomposites

CdS incorporated PANI Nano composite was Synthesis using in-situ polymerization of aniline monomer using ammonium per sulfate (APS) as the polymerizing agent in the HCl acid medium. Briefly, 1.0M HCl was mixed with 0.1M aniline and stirred for 1 h to form aniline hydrochloride. Then 5 wt% of CdS nanoparticles were added into the above solution and

vigorously stirred by magnetic stirring apparatus for 1 h at ambient temperature in order to keep the CdS homogeneously suspended in the solution. After that 0.1M APS solution was added drop wise to the above solution with continuous stirring at 5 ^{0}C for 5 h to complete the polymerization, and it was kept overnight. The outcome of resulting precipitated dark green-colored CdS incorporated PANI nano composite products was filtered and washed successively several times with deionized water until the filtrate was colorless, and then the obtained product was dried at 60°C overnight under vacuum. Fine powders of CdS incorporated PANI nanocomposites were obtained through grinding those with a mortar and pestle.

Chapter 4

4. Result and Discussion

4.1 X-ray Diffraction Studies

The crystallographic analysis of the synthesized PANI, CdS and CdS incorporated PANI nano composite powder have been investigated using X-ray diffractometer (XRD) with a scanning range of 15° to 60° using Cu-K_a radiation ((λ=1.5418 Å)) at an operating current of 40 mA and voltage 40 kV. Figure 4.1 shows the results of XRD of pure PANI, CdS and CdS incorporated PANI Nano composite materials.

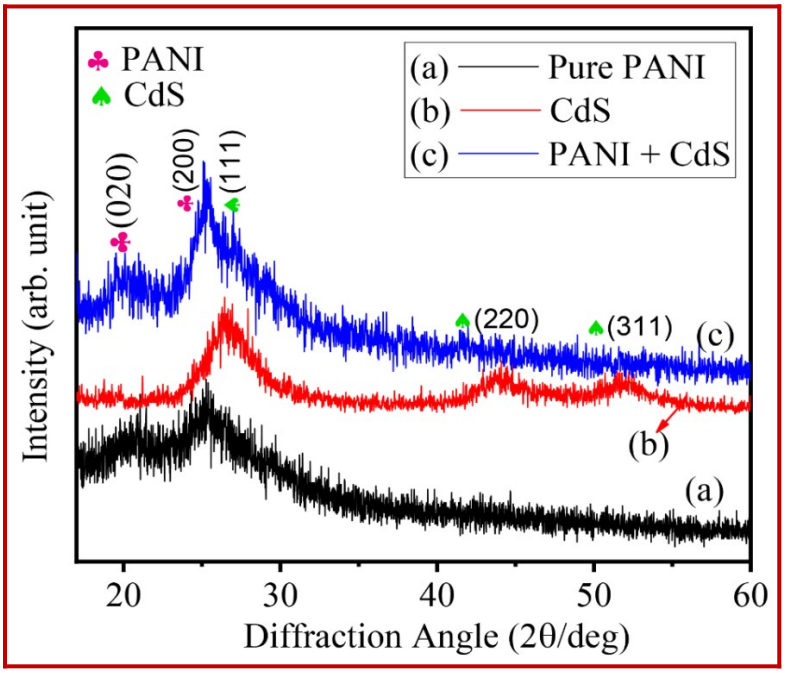


Figure 4.1 XRD patterns of (a) pure PANI, (b) CdS and (c) CdS incorporated PANI.

The crystalline phases of conductive polymers have profound significance for the subject of the enhancement of their properties. The orderly arrangement of the polymer chain becomes favorable due to crystal structure formation and this facilitates long range propagation of polaron and bipolarons states through the polymer chain. For pure PANI, two broad diffraction peaks at around 2θ = 19.7ºand 25.18º corresponding to miller plane (020) and (200) are observed. The XRD pattern of a typical CdS sample exhibits prominent broad peaks at 2θ values of 26.7º, 44º and 52º, which could be indicated as scattering from the (111), (220), and (311) cubic phase CdS plans, respectively, suggesting that the nanoparticles are in cubic (Zinc blend phase) form and are in good agreement with the reported data on CdS. For the CdS incorporated PANI composite, the intensity of the diffraction peaks for CdS are lower than that of pure CdS, due to the presence of PANI matrix. The broad and low intensity peak of PANI overlapped with the CdS (111) peak. This suggests that the crystalline CdS in the composite generate a more ordered molecular arrangement of PANI, which support to move charge carriers with higher mobility in the PANI backbone. This crystal structure behavior affects the optical and electrical properties of the composite samples

4.2 Optical Property Studies

Optical absorbance measurements can act as probes to obtain information about the different energy bands generated in PANI and its defect levels. In order to understand the effect of CdS on the fundamental optical absorption properties of PANI, optical absorption spectroscopic measurements have been observed for CdS incorporated PANI composites. Figure 4.2 shows the results of optical absorption spectra of pure PANI and CdS incorporated PANI composite. For the pure PANI, the absorption bands are centered at 410 nm and a broad hump around 850 nm. The peak observed at 410 nm is due to polaron-π* transition, while the broad hump around 850 nm is attributed to π-polaron transition, which confirm the formation of conducting phase of PANI. Furthermore, when CdS nanoparticles are dispersed in the PANI matrix, a significant change is measured in the absorption spectrum, which may be due to the interaction between the PANI and CdS in the composite, and leads to an increase in the electrical conductivity.

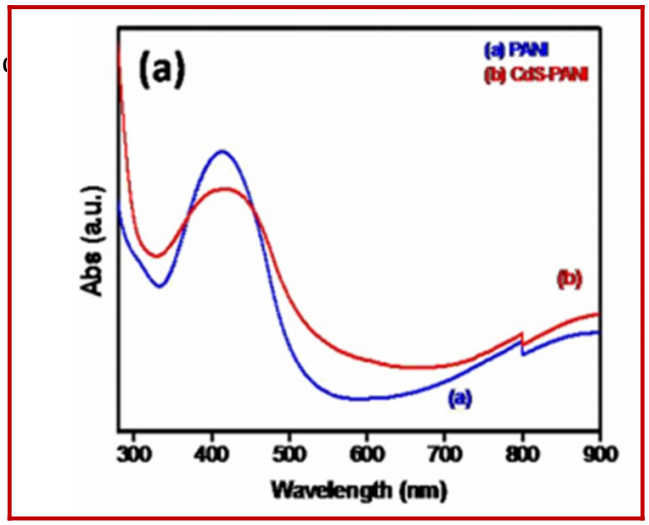


Figure 4.2 UV-vis spectra of pure PANI and CdS incorporated PANI.

4.3 Electrical Conductivity Studies

Transport properties of electrically conducting polymers involve a number of charge transport mechanisms due to their complex structural and morphological forms. Charge carriers hopping along and between polymer chains and tunneling between high-conductive crystallites embedded into amorphous matrix and electron phonon interactions play significant roles in the charge transfer mechanism of such polymers.

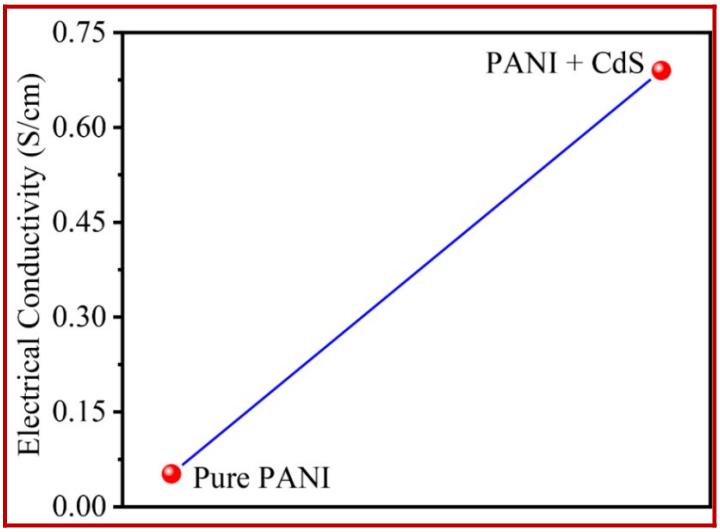


Figure 4.3 Electrical conductivity of pure PANI and CdS incorporated PANI at Room temperature.

Figure 4.3 shows the room temperature electrical conductivity of the pure PANI and CdS incorporated PANI. The value of electrical conductivity (σ) for CdS incorporated PANI composite becomes 0.69 S/cm, which is 13 times higher than that of the pure PANI (0.052 S/cm).

The enhancement of electrical conductivity may be explained to be due to the CdS nanoparticles template effect, which provides the conducting PANI to form more ordered molecular chains structure. The better ordering of the chain packing of PANI is evident from the XRD analysis. The electrical transport of PANI-based composite strongly depends on the ordering of the molecular structure. The higher degree of ordering of the chain packing influences the carrier mobility and electrical conductivity by tuning the hopping activation energy and hopping free path.

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Chapter 5

5. Conclusion and Future scope

5.1 Conclusion

This project focuses on the synthesis of polyaniline-cadmium sulphide (PANI/CdS) nanocomposites in powder form and to study the effect of CdS on its structural, optical and electrical properties. The Polyaniline-cadmium sulphide composite taken for structural, electrical and optical investigations are prepared by using in-situ polymerization of aniline monomer using ammonium per sulfate (APS) as the polymerizing agent in the HCl acid medium.

XRD pattern shows that the broad and low intensity peak of PANI overlapped with the CdS peak, this suggests that the crystalline CdS in the composite generate a more

ordered molecular arrangement of PANI. Optical studies confirmed the absorption band at 410 and 850 nm due to polaron-π* and π-polaron transition respectively which confirms the formation of the conducting phase of PANI. The electrical study depicts that the conductivity of PANI/CdS was found to be 13 times higher than pure PANI which is due to the CdS nanoparticles template effect.

5.1Applications and Future scope

Conducting polymer have been one of emerging fields in science and technology due to their mechanical flexibility, environmental stability, low manufacturing cost, simple synthetic procedure and have adequate synthesis techniques. Now-a-days conducting polymers are emerging as one of the thrust areas of experimental research. One of the interesting features of conducting polymers is that it is possible to control the electrical conductivity of these polymers from insulating to metallic by doping. They have unique electrical, optical, magnetic and chemical properties leading to the wide range of technological applications in various fields like batteries, electromagnetic interference shielding (EMI), sensors, corrosion protection coatings, microwave absorption, light emitting diodes etc. In future these organic-inorganic/metals nanocomposites can be used to study the tunning of optical and electrical properties on doping with metal or inorganic nanomaterials, and can be used in the polymer based electronics, solar cells e.t.c.